

Chapter 20

Application of Integrals

Solutions

SECTION - A

Objective Type Questions (One option is correct)

1. The area bounded by $y = -x^2 + 1$ and the x-axis is

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $\frac{4}{3}$

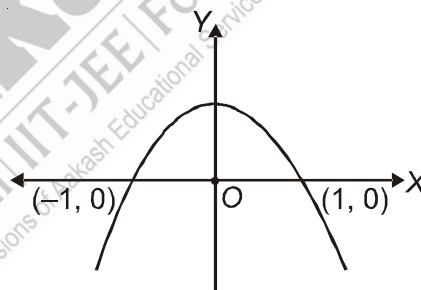
(4) $\frac{8}{3}$

Sol. Answer (3)

Required area = $2 \int_0^1 (1 - x^2) dx$

$$= 2 \left(x - \frac{x^3}{3} \right)_0^1$$

$$= 2 \left(1 - \frac{1}{3} \right) = \frac{4}{3} \text{ sq. units}$$



2. The area bounded by $y = x^2$, $x + y = 2$ is

(1) $\frac{9}{2}$

(2) $\frac{15}{2}$

(3) 9

(4) 15

Sol. Answer (1)

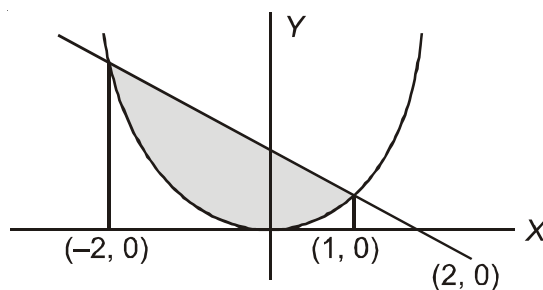
Solving $y = x^2$ and $y = 2 - x$

$$\Rightarrow x^2 = 2 - x$$

$$\Rightarrow x^2 + x - 2 = 0$$

$$\Rightarrow (x + 2)(x - 1) = 0$$

$$\Rightarrow x = -2, 1$$



$$\begin{aligned}
 \text{Required area} &= \int_{-2}^1 \{(2-x) - x^2\} dx \\
 &= \left[2x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-2}^1 \\
 &= 2(3) - \frac{1}{2}(1-4) - \frac{1}{3}(1+8) \\
 &= 6 - \frac{3}{2} - 3 \\
 &= \frac{9}{2} \text{ sq. unit}
 \end{aligned}$$

3. The area of the region bounded by the function $f(x) = x^3$, the x -axis and the lines $x = -1$ and $x = 1$ is equal to

- (1) $\frac{1}{4}$ sq. unit (2) $\frac{1}{2}$ sq. unit (3) 1 sq. unit (4) $\frac{1}{8}$ sq. unit

Sol. Answer (2)

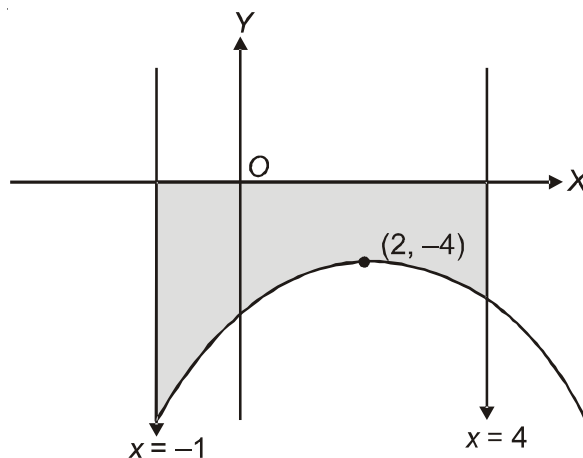
$$\begin{aligned}
 \text{Required Area} &= \left| \int_{-1}^0 x^3 dx \right| + \int_0^1 x^3 dx \\
 &= \frac{1}{4} + \frac{1}{4} \\
 &= \frac{1}{2} \text{ sq. units}
 \end{aligned}$$

4. The area of the region bounded by the x -axis, the function $y = -x^2 + 4x - 8$ and the $x = -1$ and $x = 4$, is equal to

- (1) $31\frac{2}{3}$ sq. units (2) 31 sq. units (3) $32\frac{2}{3}$ sq. units (4) 32 sq. units

Sol. Answer (1)

$$\begin{aligned}
 \text{Required area} &= \left| \int_{-1}^4 (-x^2 + 4x - 8) dx \right| \\
 &= \left| \left\{ \frac{-x^3}{3} + 2x^2 - 8x \right\}_{-1}^4 \right| \\
 &= \left| \left(\frac{-64}{3} \right) - \frac{31}{3} \right| \\
 &= \frac{95}{3} = 31\frac{2}{3} \text{ sq. units}
 \end{aligned}$$

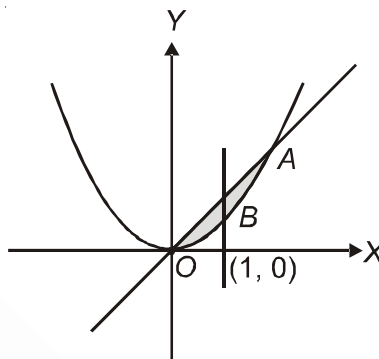


5. The area of the region bounded by $y = x^2$ and $y = 4x$, for x between 0 and 1, is equal to

- (1) 2 sq. units (2) $\frac{5}{3}$ sq. units (3) 6 sq. units (4) 1 sq. units

Sol. Answer (2)

$$\begin{aligned}\text{Required area} &= \int_0^1 (y_1 - y_2) dx \\ &= \int_0^1 (4x - x^2) dx \\ &= \left\{ 2x^2 - \frac{x^3}{3} \right\}_0^1 \\ &= \frac{5}{3} \text{ sq. units}\end{aligned}$$

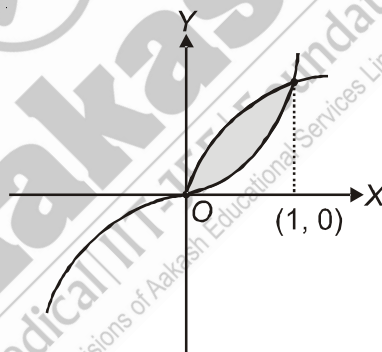


6. The area of the region in first quadrant bounded by the curves $y = x^3$ and $y = \sqrt{x}$, is equal to

- (1) $\frac{12}{5}$ sq. units (2) $\frac{5}{12}$ sq. units (3) $\frac{5}{3}$ sq. units (4) $\frac{3}{5}$ sq. units

Sol. Answer (2)

$$\begin{aligned}\text{Required area} &= \int_0^1 (\sqrt{x} - x^3) dx \\ &= \left\{ \frac{2}{3} x^{\frac{3}{2}} - \frac{x^4}{4} \right\}_0^1 \\ &= \frac{5}{12} \text{ sq. units}\end{aligned}$$

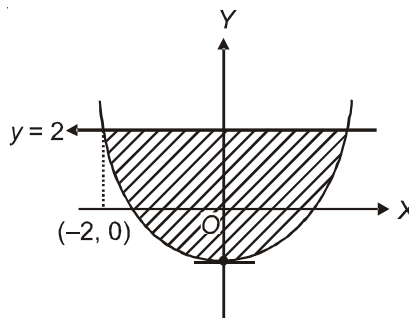


7. The area of the region bounded by the curve $y = x^2 - 2$ and line $y = 2$, is equal to

- (1) $\frac{32}{3}$ sq. unit (2) $\frac{19}{2}$ sq. unit (3) $\frac{21}{5}$ sq. unit (4) $\frac{16}{3}$ sq. unit

Sol. Answer (1)

$$\begin{aligned}\text{Required area} &= 2 \int_{-2}^2 \sqrt{y+2} dy \\ &= 2 \left\{ \frac{2}{3} (y+2)^{\frac{3}{2}} \right\}_{-2}^2 \\ &= \frac{32}{3} \text{ sq. units}\end{aligned}$$

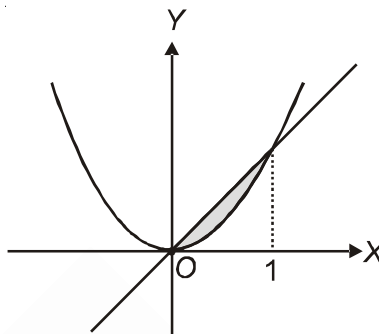


8. The area of the region bounded by the curve $y = x^2$ and $y = x$ is equal to

- (1) $\frac{9}{2}$ sq. units (2) $\frac{7}{2}$ sq. units (3) $\left(e - \frac{1}{e}\right)^2$ sq. units (4) $\frac{1}{6}$ sq. units

Sol. Answer (4)

$$\begin{aligned}\text{Required area} &= \int_0^1 (x - x^2) dx \\ &= \left\{ \frac{x^2}{2} - \frac{x^3}{3} \right\}_0^1 \\ &= \frac{1}{2} - \frac{1}{3} \\ &= \frac{1}{6} \text{ sq. units}\end{aligned}$$

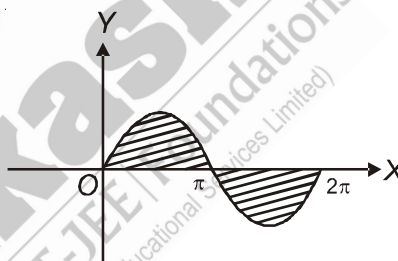


9. The area bounded by the curve $y = \sin x$, $x \in [0, 2\pi]$ and the x -axis, is equal to

- (1) 4 sq. unit (2) 0 sq. unit (3) 2 sq. unit (4) 6 sq. unit

Sol. Answer (1)

$$\begin{aligned}\text{Required area} &= \left| \int_0^{2\pi} \sin x \, dx \right| \\ &= \left| \int_0^{\pi} \sin x \, dx + \int_{\pi}^{2\pi} \sin x \, dx \right| \\ &= 2 + 2 \\ &= 4 \text{ sq. units}\end{aligned}$$

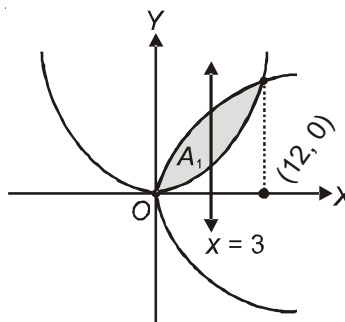


10. The ratio in which the area bounded by the curves $y^2 = 12x$ and $x^2 = 12y$ is divided by the line $x = 3$, is equal to

- (1) 15 : 16 (2) 15 : 49 (3) 1 : 2 (4) 15 : 29

Sol. Answer (2)

$$\begin{aligned}\text{Total area} &= \int_0^{12} \left(\sqrt{12x} - \frac{x^2}{12} \right) dx \\ \Rightarrow A &= \left\{ \sqrt{12} \frac{2}{3} x^{3/2} - \frac{x^3}{36} \right\}_0^{12} \\ &= \frac{2}{3} (12)^2 - \frac{(12)^3}{36} \\ &= \frac{(12)^2}{3} \\ &= 48 \text{ sq. units}\end{aligned}$$



$$\text{Now, } A_1 = \int_0^3 \left(\sqrt{12x} - \frac{x^2}{12} \right) dx$$

$$= \left\{ \frac{2}{3} \sqrt{12} x^{\frac{3}{2}} - \frac{x^3}{36} \right\}_0^3$$

$$= 12 - \frac{3}{4}$$

$$= \frac{45}{4} \text{ sq. units}$$

$$\text{So, required ratio} = \frac{A_1}{A - A_1}$$

$$= \frac{\frac{45}{4}}{48 - \frac{45}{4}}$$

$$= \frac{45}{48 \times 4 - 45}$$

$$= \frac{15}{49}$$

11. The area between the curve $y^2 = 4x$, y -axis, and $y = -1$ and $y = 3$ is equal to

(1) $\frac{7}{3}$ sq. units

(2) $\frac{9}{4}$ sq. units

(3) $\frac{1}{12}$ sq. units

(4) $\frac{1}{4}$ sq. units

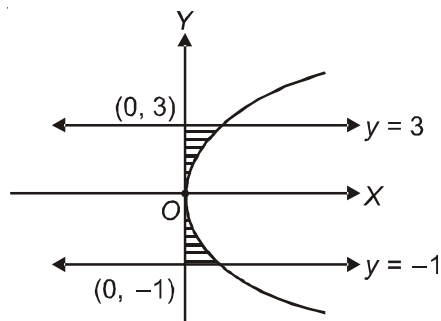
Sol. Answer (1)

$$\text{Required area} = \int_{-1}^3 \frac{y^2}{4} dy$$

$$= \left\{ \frac{y^3}{12} \right\}_{-1}^3$$

$$= \frac{27}{12} + \frac{1}{12}$$

$$= \frac{7}{3} \text{ sq. units}$$



12. The common area of the curves $y = \sqrt{x}$ and $x = \sqrt{y}$ is equal to

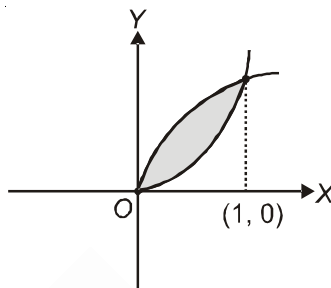
- (1) 3 sq. units (2) $\frac{5}{3}$ sq. units (3) $\frac{1}{3}$ sq. units (4) $\frac{2}{3}$ sq. units

Sol. Answer (3)

$$\text{Required area} = \int_0^1 (\sqrt{x} - x^2) dx$$

$$= \left\{ \frac{2}{3} x^{\frac{3}{2}} - \frac{x^3}{3} \right\}_0^1$$

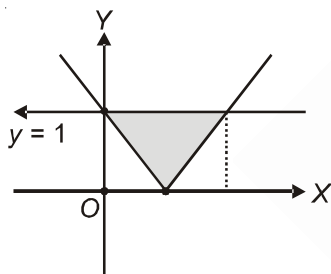
$$= \frac{1}{3} \text{ sq. units}$$



13. The area of the region bounded by $y = |x - 1|$ and $y = 1$ is equal to

- (1) 1 sq. units (2) 2 sq. units (3) $\frac{1}{2}$ sq. unit (4) 3 sq. units

Sol. Answer (1)



$$\text{Required area} = \frac{1}{2} \times 2 \times 1$$

$$= 1 \text{ sq. unit}$$

14. If area bounded by the curve $x = ay^2$ and $y = ax^2$ is 1 sq. unit, then the positive value of a is equal to

- (1) $\frac{1}{\sqrt{3}}$ (2) $\frac{1}{3}$ (3) $\frac{1}{2}$ (4) 3

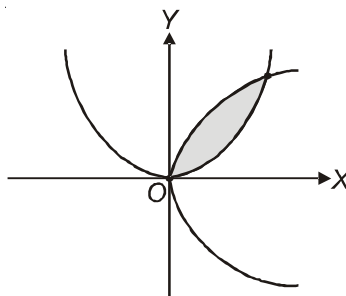
Sol. Answer (1)

Solving,

$$x = a(a^2x^4)$$

$$\Rightarrow x(1 - a^3x^3) = 0$$

$$\Rightarrow x = 0, \frac{1}{a}$$



$$\text{So, given } \int_0^{1/a} \left\{ \sqrt{\frac{x}{a}} - ax^2 \right\} dx = 1$$

$$\Rightarrow \left\{ \frac{1}{\sqrt{a}} \frac{2}{3} x^{\frac{3}{2}} - \frac{ax^3}{3} \right\}_0^{1/a} = 1$$

$$\Rightarrow \frac{2}{3a^2} - \frac{1}{3a^2} = 1$$

$$\Rightarrow \frac{1}{3a^2} = 1$$

$$\Rightarrow a^2 = \frac{1}{3}$$

$$\text{so, } a = \frac{1}{\sqrt{3}}$$

15. The area of the region bounded by the curve $y = x^2$ and $y = |x|$ is equal to

(1) $\frac{5}{3}$ sq. units

(2) $\frac{1}{3}$ sq. unit

(3) $\frac{5}{6}$ sq. units

(4) $\frac{1}{6}$ sq. unit

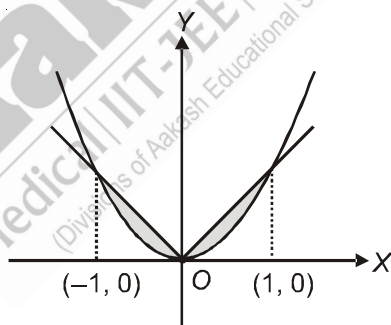
Sol. Answer (2)

$$\text{Required area} = 2 \int_0^1 (x - x^2) dx$$

$$= 2 \left\{ \frac{x^2}{2} - \frac{x^3}{3} \right\}_0^1$$

$$= 2 \left(\frac{1}{6} \right)$$

$$= \frac{1}{3} \text{ sq. units}$$



16. The slope of the tangent to the curve $y = f(x)$ at a point (x, y) is $2x + 1$ and the curve passes through the point $(1, 2)$. The area of the region bounded by the curve, the x -axis and the line $x = 1$ is equal to

(1) $\frac{5}{3}$ sq. units

(2) $\frac{5}{6}$ sq. units

(3) $\frac{6}{5}$ sq. units

(4) 6 sq. units

Sol. Answer (2)

$$\text{Given } \frac{dy}{dx} = 2x + 1 \Rightarrow y = x^2 + x + C$$

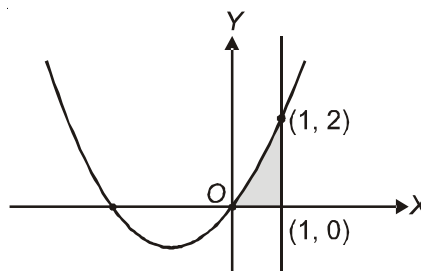
as curve passes through $(1, 2) \Rightarrow C = 0$

so, required curve $y = x^2 + x$

$$\text{Now, area} = \int_0^1 (x^2 + x) dx$$

$$= \left\{ \frac{x^3}{3} + \frac{x^2}{2} \right\}_0^1$$

$$= \frac{5}{6} \text{ sq. units}$$



17. The area of the region bounded by the curves $y = xe^x$, $y = e^x$ and the lines $x = \pm 1$, is equal to

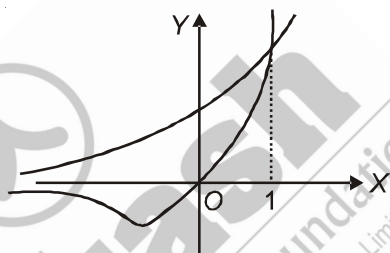
- (1) e sq. units (2) $e + \frac{2}{e}$ sq. units (3) $e - \frac{3}{e}$ sq. units (4) $e + \frac{3}{e}$ sq. units

Sol. Answer (3)

$$\text{Required area} = \int_{-1}^1 (e^x - xe^x) dx$$

$$= (e^x - e^x(x-1))_{-1}^1$$

$$= \left(e - \frac{3}{e} \right) \text{ sq. units}$$



18. The area between the curves $y = x^3$ and $y = x + |x|$ is equal to

- (1) 0 sq. units (2) 2 sq. units (3) 1 sq. units (4) 3 sq. units

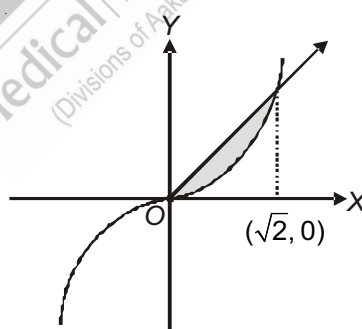
Sol. Answer (3)

$$\text{Required area} = \int_0^{\sqrt{2}} (2x - x^3) dx$$

$$= \left\{ x^2 - \frac{x^4}{4} \right\}_0^{\sqrt{2}}$$

$$= \left(2 - \frac{4}{2} \right) - 0$$

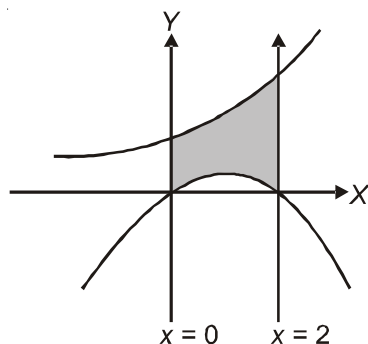
$$= 1 \text{ sq. unit}$$



19. Area bounded by the curves $y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = 2$ is equal to

- (1) $\frac{3}{\log 2} - \frac{4}{3}$ sq. units (2) $\frac{3}{\log 2} + \frac{4}{3}$ sq. units

- (3) $3\log 2 + \frac{4}{3}$ sq. units (4) $3\log 2 - \frac{4}{3}$ sq. units

Sol. Answer (1)

$$\begin{aligned}\text{Required area} &= \int_0^2 \{2^x - (2x - x^2)\} dx = \left\{ \frac{2^x}{\log 2} - x^2 + \frac{x^3}{3} \right\}_0^2 \\ &= \left(\frac{4}{\log 2} - 4 + \frac{8}{3} \right) - \left(\frac{1}{\log 2} \right) = \left(\frac{3}{\log 2} - \frac{4}{3} \right) \text{ sq. units}\end{aligned}$$

20. For which of the following positive value of m , the area of the region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals $\frac{9}{2}$?

(1) 3

(2) 1

(3) 2

(4) 4

Sol. Answer (4)

Solving the curve $y = x - x^2$ with the line $y = mx$, we have $x = 0$ and $x = 1 - m$.

$$\text{so, } \left| \int_0^{1-m} \{(x - x^2) - mx\} dx \right| = \frac{9}{2} \text{ gives}$$

$$\Rightarrow \left\{ (1-m) \frac{x^2}{2} - \frac{x^3}{3} \right\}_0^{1-m} = \pm \frac{9}{2}$$

$$\Rightarrow \frac{(1-m)^3}{2} - \frac{(1-m)^3}{3} = \pm \frac{9}{2}$$

$$\Rightarrow \frac{(1-m)^3}{6} = \pm \frac{9}{2}$$

$$\Rightarrow 1 - m = \pm 3$$

$$\Rightarrow m = -2, 4$$

so, positive value of ' m ' = 4.

21. The area bounded by the curve $y = (x - 1)^2$, $y = (x + 1)^2$ and the x-axis is

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{4}{3}$ (4) $\frac{8}{3}$

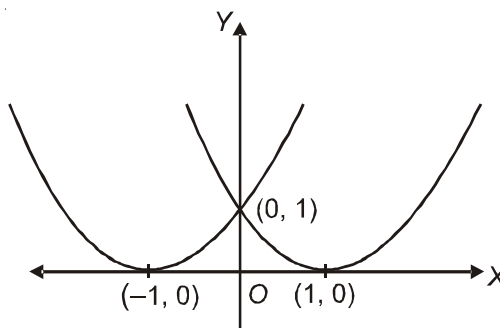
Sol. Answer (2)

The required area = $2 \int_0^1 (x-1)^2 dx$

$$= \frac{2}{3} \left[(x-1)^3 \right]_0^1$$

$$= \frac{2}{3} (0+1)$$

$$= \frac{2}{3}$$



22. The smaller area bounded by $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the line $3x + 4y = 12$ is

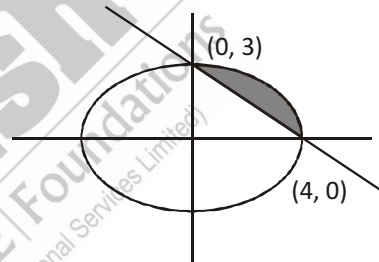
(1) 3π sq. units(2) $(3\pi - 6)$ sq. units(3) $(3\pi - 2)$ sq. units(4) $(3\pi - 4)$ sq. units**Sol.** Answer (2)

$$A = \int_0^4 \left\{ \frac{3}{4} \sqrt{16-x^2} - \left(3 - \frac{3x}{4} \right) \right\} dx$$

$$= \frac{3}{4} \left\{ \frac{x}{2} \sqrt{16-x^2} + \frac{16}{2} \sin^{-1} \frac{x}{4} \right\}_0^4 - \frac{3}{4} \left\{ 4x - \frac{x^2}{2} \right\}_0^4$$

$$= \frac{3}{4} \left\{ \left(0 + \frac{8\pi}{2} \right) - (0+0) \right\} - \frac{3}{4} \left\{ 4 \times 4 - \frac{1}{2} \times 16 \right\}$$

$$= (3\pi - 6) \text{ sq. units}$$



23. The area bounded by the curve $y = |x| - 1$ and $y = -|x| + 1$ is

(1) 1 sq. unit

(2) 2 sq. unit

(3) $2\sqrt{2}$ sq. unit

(4) 4 sq. unit

Sol. Answer (2)

Area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$

Area required

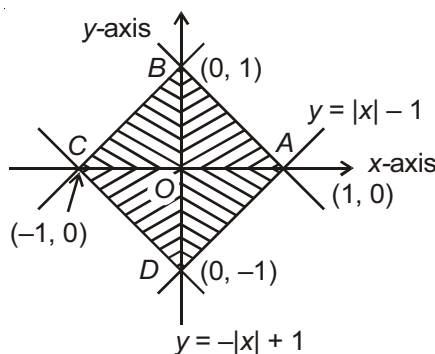
$$= 4 \text{ Area of } \triangle OAB$$

$$= 4 \times \frac{1}{2} \times \text{base} \times \text{height}$$

$$= 4 \times \frac{1}{2} \times 1 \times 1$$

$$= 2 \text{ square units.}$$

Area bounded by $y = |x| - 1$ and $y = -|x| + 1 = 2$ square unit.



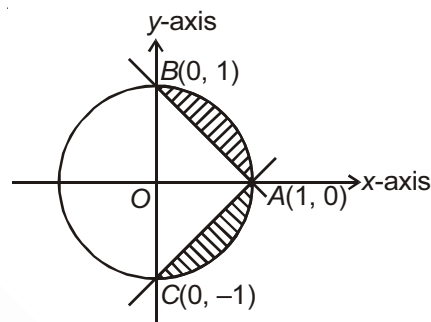
24. The area bounded by curve $|x| + |y| \geq 1$ and $x^2 + y^2 \leq 1$ for $x \geq 0$ is

- (1) 2 sq. unit (2) $\frac{\pi}{2}$ sq. unit (3) $\frac{(\pi-2)}{2}$ sq. unit (4) $(\pi-2)$ sq. unit

Sol. Answer (3)

The area bounded by the curve $|x| + |y| \geq 1$ and $x^2 + y^2 \leq 1$ for $x \geq 0$ is

$$\begin{aligned} \text{Required Area} &= 2 \left[\int_0^1 y_{\text{circle}} \right] - 2 (\text{area of } \triangle AOB) \\ &= 2 \left[\int_0^1 \sqrt{1-x^2} dx \right] - 2 \cdot \frac{1}{2} \times \text{base} \times \text{height} \\ &= 2 \left[\frac{x\sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1} x \right]_0^1 - \left(2 \times \frac{1}{2} \times 1 \times 1 \right) \\ &= 2 \cdot \frac{1}{2} \sin^{-1} 1 - 1 \\ &= \frac{\pi}{2} - 1 = \left(\frac{\pi-2}{2} \right) \text{ square units.} \end{aligned}$$



Second Method

Required area = semi-circle - $\triangle ABC$

$$\begin{aligned} &= \frac{1}{2} \pi - \frac{1}{2} \cdot 2 \cdot 1 \\ &= \left(\frac{\pi-2}{2} \right) \text{ square units.} \end{aligned}$$

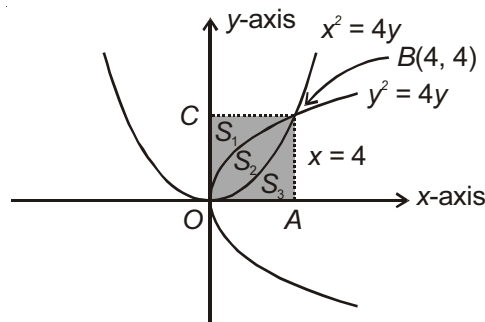
25. The parabola $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4$, $y = 4$ and the coordinate axes of S_1 , S_2 , S_3 are respectively the areas of these parts numbered from top to bottom, then $S_1 : S_2 : S_3$ is

- (1) 1 : 1 : 1 (2) 2 : 1 : 2 (3) 1 : 2 : 3 (4) 1 : 2 : 1

Sol. Answer (1)

For point of intersection of the given parabolas $x^2 = 4y$ and $y^2 = 4x$, we have

$$\begin{aligned} x^4 &= 16y^2 \\ \Rightarrow x^4 &= 16 \cdot 4x \\ \Rightarrow x(x^3 - 64) &= 0 \\ \Rightarrow x &= 0 \text{ or } x = 4 \\ \therefore \text{Points of intersection are } O(0, 0) \text{ and } B(4, 4) \end{aligned}$$



$$\begin{aligned}\text{Area of } S_3 &= \int_0^4 \frac{x^2}{4} dx = \left[\frac{x^3}{12} \right]_0^4 \\ &= \frac{64}{12} = \frac{16}{3} \text{ square units.}\end{aligned}$$

$$\begin{aligned}\text{Area of } S_2 &= \int_0^4 2\sqrt{x} dx - \frac{16}{3} \\ &= \frac{4}{3} [x^{3/2}]_0^4 - \frac{16}{3} \\ &= \frac{32}{3} - \frac{16}{3} = \frac{16}{3} \text{ square units.}\end{aligned}$$

$$\text{Area of } S_1 = \text{Area of square} - \text{Area of } S_1 - \text{Area of } S_2$$

$$= 16 - \frac{16}{3} - \frac{16}{3} = \frac{16}{3} \text{ square units.}$$

$$S_1 : S_2 : S_3 = 1 : 1 : 1$$

26. Area bounded by the curves $y = x^2 - 1$, $x + y = 3$ is

(1) $\frac{9}{2}$

(2) 4

(3) $\frac{7\sqrt{17}}{2}$

(4) $\frac{17\sqrt{17}}{6}$

Sol. Answer (4)

$$\text{Curves are } y = x^2 - 1 \quad \dots (i)$$

$$x + y = 3 \quad \dots (ii)$$

Solving (i) and (ii) for point of intersection, we have

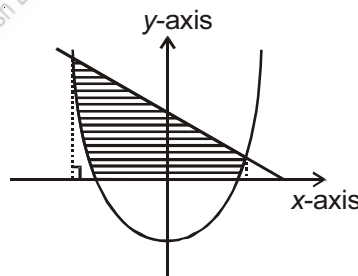
$$x^2 - 1 = 3 - x$$

$$x^2 + x - 4 = 0$$

$$x = \frac{-1 \pm \sqrt{17}}{2} = \frac{-1 + \sqrt{17}}{2}, \frac{-1 - \sqrt{17}}{2}$$

$$\text{Required area} = \int_{\frac{-1-\sqrt{17}}{2}}^{\frac{-1+\sqrt{17}}{2}} ((3-x) - (x^2-1)) dx = \int_{\frac{-1-\sqrt{17}}{2}}^{\frac{-1+\sqrt{17}}{2}} (4-x-x^2) dx$$

$$\begin{aligned}&= \left[4x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{\frac{-1-\sqrt{17}}{2}}^{\frac{-1+\sqrt{17}}{2}} \\ &= \frac{17\sqrt{17}}{6}\end{aligned}$$



27. The area bounded between curves $y^2 = x$ and $y = |x|$ is

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) 1

(4) $\frac{1}{6}$

Sol. Answer (4)

Area bounded by curves $y^2 = x$ and $y = |x|$

For point of intersection

$$|x| = x \Rightarrow x^2 = x$$

$$x^2 - x = 0$$

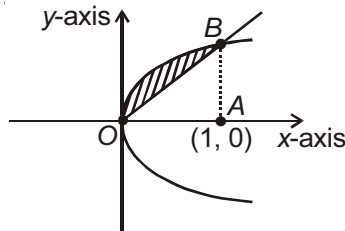
$$x = 0$$

$$x = 1$$

The point of intersections are $(0, 0)$, $(1, 1)$

$$\text{Required Area} = \int_0^1 \sqrt{x} dx - \int_0^1 x dx$$

$$= \frac{1}{6} \text{ square unit.}$$



28. The area of figure bounded by the curves $y = a^x$ ($a > 1$) and $y = b^{-x}$ ($b > 1$) and the straight line $x = 1$ is

(1) $\frac{1}{\log a}(a-1) + \frac{1}{\log b}\left(\frac{1}{b}-1\right)$

(2) $\log a(a-1) + \log b\left(\frac{1}{b}-1\right)$

(3) $\frac{1}{\log a}(a-1) + \frac{1}{\log b}(b-1)$

(4) $\log a(a-1) + \log b \cdot (b-1)$

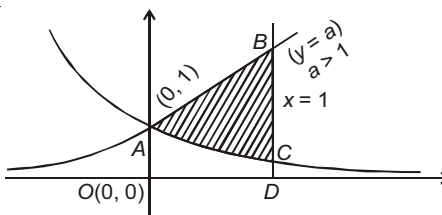
Sol. Answer (1)

Area of figure bounded by the curve $y = a^x$ ($a > 1$), and $y = b^{-x}$ ($b > 1$) and the straight line $x = 1$

$$\text{Required Area} = \int_0^1 (a^x - b^{-x}) dx$$

$$= \left[\frac{a^x}{\log a} \right]_0^1 + \left[\frac{b^{-x}}{\log b} \right]_0^1$$

$$= \frac{1}{\log a}[a-1] + \frac{1}{\log b}\left[\frac{1}{b}-1\right].$$



29. Minimum area of triangle formed by any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with the co-ordinate axes is

(1) $\frac{a^2 + b^2}{2}$

(2) $\frac{(a+b)^2}{2}$

(3) ab

(4) $\frac{(a-b)^2}{2}$

Sol. Answer (3)

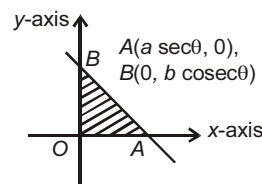
Let any point on the given ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be $(a \cos \theta, b \sin \theta)$

then tangent to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at point $(a \cos \theta, b \sin \theta)$ is $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$

then point of intersection on co-ordinate axis by this tangent are $(a \sec \theta, 0)$, $(0, b \operatorname{cosec} \theta)$, the area of

$$\Delta OAB = \frac{1}{2}(OA)(OB) = \frac{1}{2} \times \text{base} \times \text{height}$$

$$A = \frac{1}{2} ab \sec \theta \operatorname{cosec} \theta = \frac{1}{2} \frac{ab}{\sin \theta \cos \theta} = ab \operatorname{cosec} 2\theta$$



$\therefore$ Minimum value of $\operatorname{cosec} 2\theta$ is 1 which occurs when $\theta = \frac{\pi}{4}$

$\therefore$ Minimum area = ab square units.

30. The area enclosed by the curve $y^2 + x^4 = x^2$ is

- (1) $\frac{2}{3}$ (2) $\frac{4}{3}$ (3) $\frac{8}{3}$ (4) $\frac{16}{3}$

Sol. Answer (2)

We have, $y^2 = x^2 - x^4 = x^2(1 - x^2)$

$$\text{Area} = \left| 4 \int_{-1}^0 x \sqrt{1-x^2} dx \right|$$

Put $t = 1 - x^2$, then

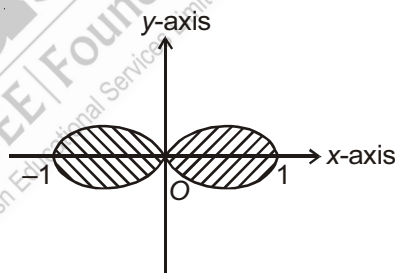
$$-\frac{dt}{2} = x dx$$

When $x = -1$, $t = 1 - 1 = 0$

When $x = 0$, $t = 1$

$$= 4 \left[-\frac{1}{2} \int_0^1 \sqrt{t} dt \right]$$

$$= \frac{4}{2} \left| \int_0^1 \sqrt{t} dt \right| = 2 \cdot \left[\frac{t^{3/2}}{3/2} \right]_0^1 = \frac{4}{3} [1 - 0] = \frac{4}{3}$$



Thus area enclosed by the curve $y^2 + x^4 = x^2$ is $\frac{4}{3}$ square units.

31. If the area bounded by x-axis, the curve $y = f(x)$ and the ordinates $x = c$ and $x = d$ is independent of $d, \forall d > c$ (c is constant), then f is

- (1) A non-zero constant function (2) Identity function
(3) Zero function (4) Parabolic function

Sol. Answer (3)

$$\text{Required area} = \int_c^d y \, dx = \int_c^d f(x) \, dx$$

which is independent of d , it is possible only when $f(x) = 0$

32. The area bounded by curve

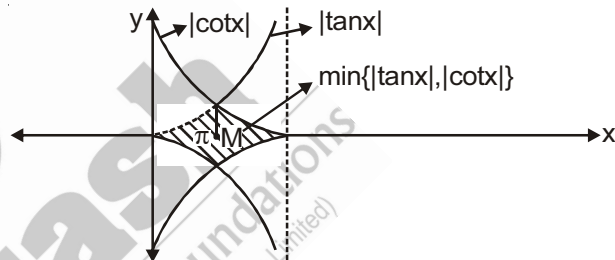
$$f(x) = \|\tan x + \cot x\| - \|\tan x - \cot x\| \text{ between lines } x=0, x=\frac{\pi}{2} \text{ and } x\text{-axis, is}$$

- (1) $\log 4$ (2) $\log \sqrt{2}$ (3) $2 \log 4$ (4) $\sqrt{2} \log 2$

Sol. Answer (1)

$$f(x) = \|\tan x + \cot x\| - \|\tan x - \cot x\| = 2 \min\{|\tan x|, |\cot x|\}$$

$$\begin{aligned} \therefore \text{Required area} &= 2 \int_0^{\pi/4} \tan x \, dx + 2 \int_{\pi/4}^{\pi/2} \cot x \, dx \\ &= 2 \left\{ \log \sqrt{2} - \log \frac{1}{\sqrt{2}} \right\} = 2 \log 2 \\ &= \log 4 \end{aligned}$$

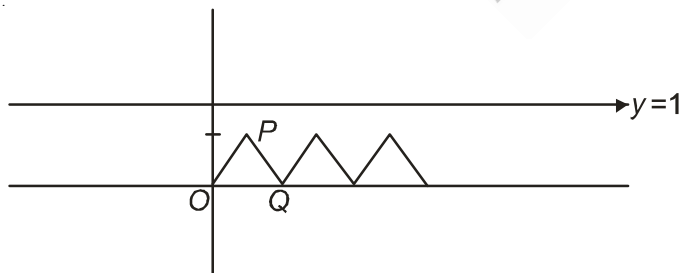


33. The area of $\int_{-10}^{10} f(x) \, dx$, where

$$f(x) = \min\{x - [x], -x - [-x]\}, \text{ where } [] \text{ is G.I.F., is}$$

- (1) 20 (2) 40 (3) 5 (4) 30

Sol. Answer (3)



$$\therefore \text{Area } (\Delta OPQ) = \frac{1}{2} \times 1 \times \frac{1}{2} = \frac{1}{4}$$

$$\text{Total area} = 20 \times \frac{1}{4} = 5$$

34. The area bounded by the curve

$$\left| \frac{x}{9} \right| + \left| \frac{y}{9} \right| = \log_b a \times \log_a b, \text{ where } a, b > 0, a \neq b \neq 1 \text{ is}$$

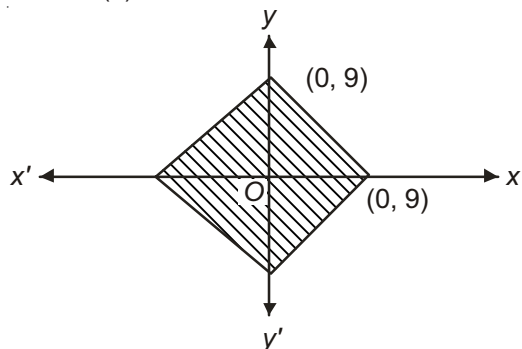
(1) 81

(2) 27

(3) 162

(4) 36

Sol. Answer (3)



Given equation is $\left| \frac{x}{9} \right| + \left| \frac{y}{9} \right| = 1$

$$\Rightarrow |x| + |y| = 9$$

$$\text{Required Area} = (9\sqrt{2})^2$$

$$= 81 \times 2$$

$$= 162 \text{ sq. units}$$

35. The area bounded by the curve $y = (x-1)(x-2)(x-3)$ and x-axis lying between the ordinates $x = 0$ and $x = 4$ is

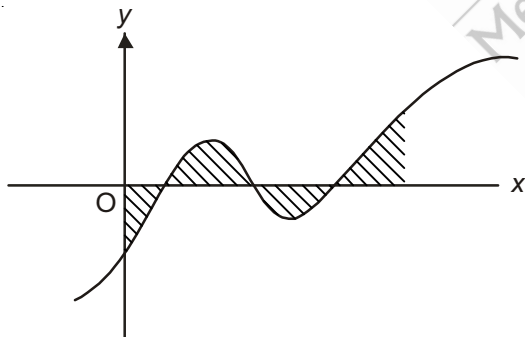
(1) $\frac{7}{4}$

(2) $\frac{9}{4}$

(3) 5

(4) $\frac{11}{4}$

Sol. Answer (3)



$$\text{Required Area} = \int_0^4 (x-1)(x-2)(x-3) dx$$

$$= \int_{-2}^2 (t+1)t + (t-1)dt \quad \text{Put } x-2 = t$$

$$= 0$$

36. The area bounded by the curve $y = \cos^{-1}(\cos x)$ and $y = |x - \pi|$ is

- (1) π^2 sq. unit (2) $2\pi^2$ sq. unit (3) $\frac{\pi^2}{2}$ sq. unit (4) $\frac{\pi^2}{4}$ sq. unit

Sol. Answer (3)

Given $y = \cos^{-1}(\cos x)$

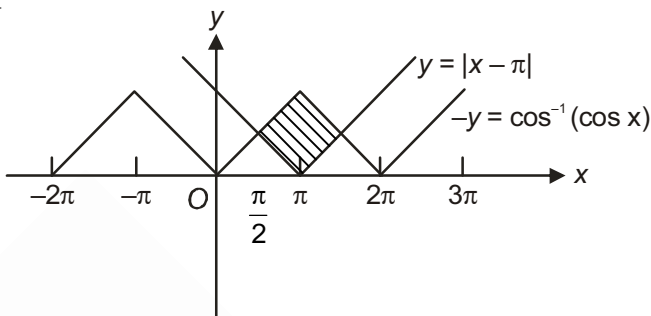
$$y = \begin{cases} x, & x \in [0, \pi] \\ 2\pi - x, & x \in [\pi, 2\pi] \end{cases}$$

Required Area = $2 \int_{\pi/2}^{\pi} (x - (\pi - x)) dx$

$$= 2(x^2 - \pi x) \Big|_{\pi/2}^{\pi}$$

$$= 2 \left(\pi^2 - \pi^2 - \frac{\pi^2}{4} + \frac{\pi^2}{2} \right)$$

$$= \frac{\pi^2}{2} \text{ sq units}$$



SECTION - B

Objective Type Questions (More than one options are correct)

1. Maximum area of the rectangle whose two vertices lie on the x-axis and two on the curve $y = 3 - |x|$ and $|x| \leq 3$ is

- (1) 9 (2) $\frac{9}{2}$ (3) $\frac{9}{4}$ (4) Less than 5

Sol. Answer (2, 4)

$$y = 3 - |x|$$

$$y = \begin{cases} 3 + x & \text{if } x < 0 \\ 3 - x & \text{if } x \geq 0 \end{cases}$$

$$\text{Area} = 2x(3 - x) \Rightarrow A'(x) = 2(3 - x) - 2x = 6 - 4x$$

$$A'(x) = 0 \Rightarrow x = \frac{3}{2} \therefore \text{Maximum area occurs when } x = \frac{3}{2}$$

$$\therefore \text{Maximum area} = 2 \left(\frac{3}{2} \right) \left(3 - \frac{3}{2} \right) = \frac{9}{2} \text{ sq. unit}$$

2. The line $y = mx$ bisects the area enclosed by the lines $x = 0$, $y = 0$ and $x = \frac{3}{2}$ and the curve $y = -x^2 + 4x + 1$, then the value of m is equal to

- (1) $\frac{13}{6}$ (2) Greater than 2 (3) Less than 1 (4) 2

Sol. Answer (1, 2)

The equation $y = 1 + 4x - x^2$ can be written as $(x - 2)^2 = -(y - 5)$

This is a downward parabola with vertex (2, 5)

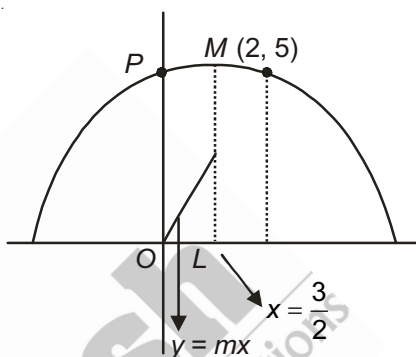
By the given condition area $OLMP = 2$ area of $\triangle OQL$

$$\Rightarrow \int_0^{3/2} (1 + 4x - x^2) dx = 2 \left(\frac{1}{2} \cdot \frac{3}{2} \cdot QL \right)$$

$$\Rightarrow \left[x + 2x^2 - \frac{x^3}{3} \right]_0^{3/2} = 2 \left(\frac{1}{2} \cdot \frac{3}{2} \cdot \frac{3m}{2} \right)$$

$$\Rightarrow \frac{3}{2} + \frac{9}{2} - \frac{27}{24} = \frac{9}{4}m$$

$$\Rightarrow \frac{9}{4}m = \frac{39}{8} \Rightarrow m = \frac{39}{9 \times 2} = \frac{13}{6}$$



3. The area bounded by $y = [x]$, $y = -[x]$, $x = 4$, where $[]$ is G.I.F., on the right side of the y -axis, is

- (1) 12 sq. unit

- (2) Equal to $18 \int_0^1 [2 \sin \pi x] dx$

- (3) Equal to area of a circle of radius $\sqrt{\frac{12}{\pi}}$

- (4) 18 sq. units

Sol. Answer (1, 2, 3)

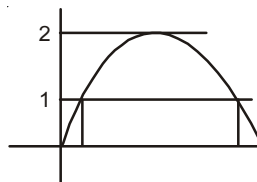
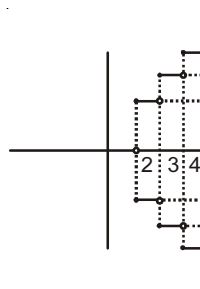
Required area = Area of 12 squares of unit length
= 12

$$\int_0^1 [2 \sin \pi x] dx$$

$$= \int_0^{1/6} 0 dx + \int_{1/6}^{1/2} 1 dx + \int_{1/2}^{5/6} 1 dx + \int_{5/6}^1 0 dx$$

$$= \frac{4}{6} = \frac{2}{3}$$

$$\therefore 18 \int_0^1 [2 \sin \pi x] dx = 12$$



SECTION - C

Linked Comprehension Type Questions

Comprehension

Computing area with parametrically represented boundaries

If the boundary of a figure is represented by parametric equations $x = x(t)$, $y = y(t)$, then the area of the figure is evaluated by one of the three formulae

$$S = \int_{\alpha}^{\beta} y(t) x'(t) dt ; S = - \int_{\alpha}^{\beta} x(t) y'(t) dt$$

$$S = \frac{1}{2} \int_{\alpha}^{\beta} (yx' - xy') dt$$

where α and β are the values of the parameter t corresponding respectively to the beginning and the end of traversal of the contour.

1. The area of ellipse enclosed by $x = a \cos t$,
 $y = b \sin t$ ($0 \leq t \leq 2\pi$)

- (1) ab (2) πab (3) $\frac{\pi}{2} ab$ (4) $2\pi ab$

Sol. Answer (2)

Lets compute $xy' - yx'$

$$\begin{aligned} xy' - yx' &= a \cos t \times b \cos t + b \sin t \times a \sin t \\ &= ab \end{aligned}$$

$$S = \frac{1}{2} \int_0^{2\pi} (xy' - yx') dt = \frac{1}{2} \int_0^{2\pi} ab dt = \pi ab$$

2. The area enclosed by the astroid $\left(\frac{x}{a}\right)^{\frac{2}{3}} + \left(\frac{y}{a}\right)^{\frac{2}{3}} = 1$ is

- (1) $\frac{a^2\pi}{8}$ (2) $\frac{a^2\pi}{4}$ (3) $\frac{3\pi a^2}{8}$ (4) $\frac{\pi a^2}{2}$

Sol. Answer (3)

Let us re-write the equation in parametric

$$\text{form } x = a \cos^3 t, y = a \sin^3 t, 0 \leq t \leq 2\pi$$

$$xy' - yx' = a^2 (\cos^3 t \cdot 3 \sin^2 t \cos t + \sin^3 t \cdot 3 \cos^2 t \sin t) = 3a^2 \cos^2 t \sin^2 t$$

$$S = \frac{1}{2} \int_0^{2\pi} (xy' - yx') dt = \frac{3}{8} a^2 \int_0^{2\pi} \sin^2 2t dt = \frac{3}{8} a^2 \pi$$

3. The area of the region bounded by an arc of cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$ and the x -axis

(1) πa^2

(2) π

(3) $3\pi a^2$

(4) $4\pi a^2$

Sol. Answer (3)

$$S = \int_{\alpha}^{\beta} yx' dt$$

$$S = - \int_{2\pi}^0 a(1 - \cos t) a(1 - \cos t) dt$$

$$= a^2 \int_0^{2\pi} (1 - \cos t)^2 dt = 3\pi a^2$$

SECTION - D

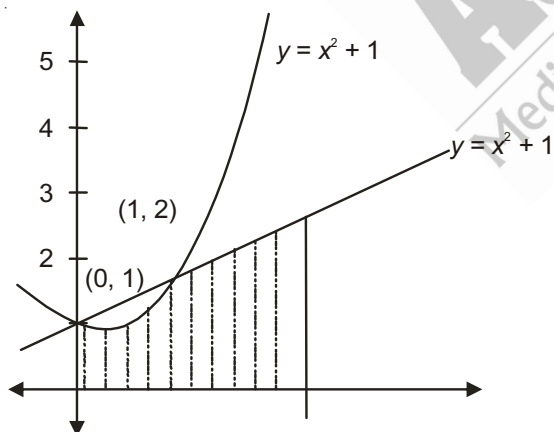
Assertion-Reason Type Questions

1. STATEMENT-1 : The area bounded by the region $\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$ is $\frac{23}{6}$ sq. units.

and

STATEMENT-2 : Required Area is $\int_a^b (y_2 - y_1) dx$

Sol. Answer (1)



$$R = \{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$$

$$= \{(x, y) : 0 \leq y \leq x^2 + 1\} \cap \{(x, y) : 0 \leq y \leq x + 1\} \cap \{(x, y) : 0 \leq x \leq 2\}$$

$$R_1 \cap R_2 \cap R_3$$

Where $R_1 = \{(x, y) : 0 \leq y \leq x^2 + 1\}$

$$R_2 = \{(x, y) : 0 \leq y \leq x + 1\}$$

$$R_3 = \{(x, y) : 0 \leq x \leq 2\}$$

Thus the sketch of R_1 , R_2 and R_3 are:

$$\text{Area} = \int_0^1 (x^2 + 1) dx + \int_1^2 (x + 1) dx = \frac{23}{6} \text{ sq. units}$$

2. STATEMENT-1 : The area bounded by the curves $y = x^2$ and $y = \frac{2}{1+x^2}$ is $2\left(\frac{\pi}{2} - \frac{1}{3}\right)$ sq. units

and

STATEMENT-2 : The curves meet at a point whose abscissae are ± 2 .

Sol. Answer (3)

For intersection

$$x^2 = \frac{2}{1+x^2}$$

$$\Rightarrow x^4 + x^2 - 2 = 0$$

$$\Rightarrow x = \pm 1$$

$$\text{Hence the required area} = 2 \int_0^1 \left(\frac{2}{1+x^2} - x^2 \right) dx = 2 \left(\frac{\pi}{2} - \frac{1}{3} \right)$$

SECTION - E

Matrix-Match Type Question

1. Area under the curve

Column-I

(A) $y = x^2$, $y = |2 - x^2|$, $y = 2$ at right of line $x = 1$

(B) $x^2 + y^2 = 25$, $4y = |4 - x^2|$, $x = 0$ and $y \geq 0$

(C) $y = x^2$, $y = \frac{2}{1+x^2}$

(D) $x^2 + y^2 = 10x$, $y = \sqrt{5x}$

Column-II

(p) $\pi - \frac{2}{3}$

(q) $25 \sin^{-1} \frac{4}{5} + 4$

(r) $25 \left(\frac{\pi}{4} - \frac{2}{3} \right)$

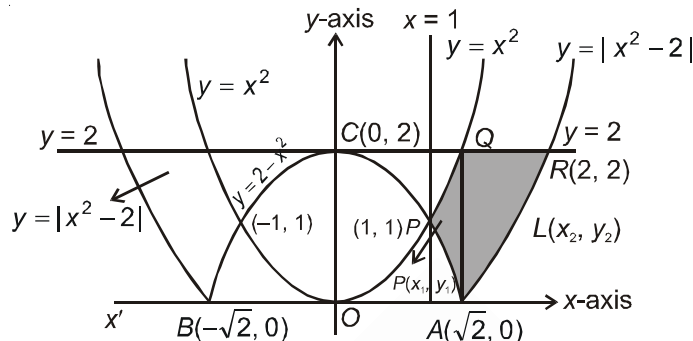
(s) $\frac{20 - 12\sqrt{2}}{3}$

Sol. Answer A(s), B(q), C(p), D(r)

Area under the curve

(A) $y = x^2$, $y = |2 - x^2|$, $y = 2$ at right of the line $x = 1$

The region bounded by the given curves on the right side of $x = 1$ is shaded in figure. Let us slice the shaded region into vertical strips.



Required area = Area of region APQA + Area of region AQRA

$$\begin{aligned}
 &= \int_1^{\sqrt{2}} (y_2 - y_1) dx + \int_{\sqrt{2}}^2 (2 - y) dx \\
 &= \int_1^{\sqrt{2}} \{x^2 - (2 - x^2)\} dx + \int_{\sqrt{2}}^2 (4 - x^2) dx \\
 &= \int_1^{\sqrt{2}} (2x^2 - 2) dx + \int_{\sqrt{2}}^2 (4 - x^2) dx \\
 &= \left[\frac{2}{3} x^3 - 2x \right]_1^{\sqrt{2}} + \left[4x - \frac{x^3}{3} \right]_{\sqrt{2}}^2 \\
 &= \left(\frac{20 - 12\sqrt{2}}{3} \right) \text{ square units.} \\
 &= \frac{4}{3} (5 - 3\sqrt{2}) \text{ sq. units.}
 \end{aligned}$$

(B) Area bounded by the curves $x^2 + y^2 = 25$, $4y = |4 - x^2|$, $x = 0$ and $y \geq 0$ and $y \geq 0$.

The equation $x^2 + y^2 = 25$ represents a circle of radius 5 with centre at (0, 0)

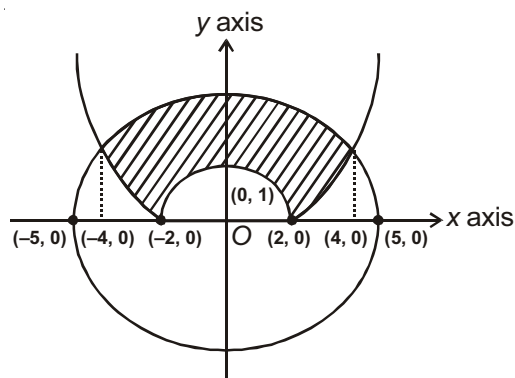
The second curve is $y = \left| \frac{4 - x^2}{4} \right| = \left| 1 - \frac{x^2}{4} \right|$

Which can be traced easily by graph trans-formation.

When the two curve intersect each other then

$$x^2 + \left(1 - \frac{x^2}{4} \right)^2 = 25$$

$$x = \pm 4$$



$$\begin{aligned}
 \text{Hence the required area} &= 2 \int_0^4 \left(\sqrt{25-x^2} - \left| 1 - \frac{x^2}{4} \right| \right) dx \\
 &= 2 \left[\int_0^4 \sqrt{25-x^2} dx - \int_0^2 \left(1 - \frac{x^2}{4} \right) dx + \int_2^4 \left(1 - \frac{x^2}{4} \right) dx \right] \\
 &= 25 \sin^{-1} \left(\frac{4}{5} \right) + 4.
 \end{aligned}$$

(C) The area of region bounded by the curves.

$$y = x^2 \text{ and } y = \frac{2}{1+x^2}$$

$$\text{For intersection point, } x^2 = \frac{2}{1+x^2}$$

$$\text{i.e., } x^4 + x^2 - 2 = 0$$

$$(x^2 + 2)(x^2 - 1) = 0$$

$$x = \pm 1$$

Hence the required area

$$= 2 \int_0^1 \left(\frac{2}{1+x^2} - x^2 \right) dx$$

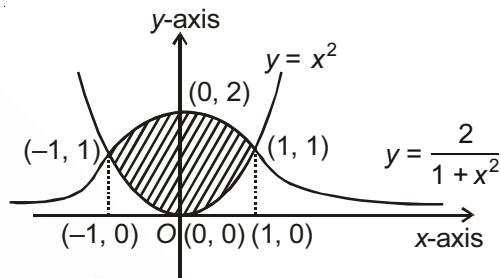
$$= 2 \left[2 \tan^{-1} x - \frac{x^3}{3} \right]_0^1$$

$$= 2 \left[\left(2 \tan^{-1} 1 - \frac{1}{3} \right) - (2 \tan^{-1} 0 - 0) \right]$$

$$= 2 \left[2 \tan^{-1} 1 - \frac{1}{3} \right]$$

$$= 2 \left[2 \left(\frac{\pi}{4} \right) - \frac{1}{3} \right] = 2 \left(\frac{\pi}{2} - \frac{1}{3} \right)$$

$$= \pi - \frac{2}{3}$$



(D) Area bounded by the curve

$$x^2 + y^2 = 10x, \quad y^2 = 5x$$

$$x^2 + y^2 - 10x = 0 \text{ is a circle centred at } (5, 0) \text{ and radius} = 5.$$

i.e., $C(5, 0)$, radius = $\sqrt{g^2 + f^2 - c} = \sqrt{25} = 5$

For point of intersection

$$10x - x^2 = 5x$$

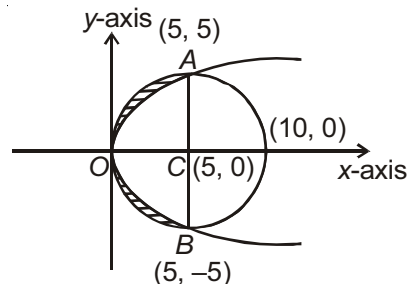
$$x^2 + 5x - 10x = 0$$

$$x^2 - 5x = 0$$

$$x = 0$$

$$x = 5$$

$\therefore$ Point of intersection are $(5, 5)$, $(5, -5)$



$$\text{Required Area} = 2 \int_0^5 (y_{\text{circle}} - y_{\text{parabola}}) dx$$

$$= 2 \left[\int_0^5 \sqrt{10x - x^2} dx - \int_0^5 \sqrt{5} \sqrt{x} dx \right] = 2 \left[\int_0^5 \sqrt{25 - (x-5)^2} dx - \sqrt{5} \left[\frac{x^{3/2}}{3/2} \right]_0^5 \right]$$

$$= 2 \left[\frac{(x-5)\sqrt{10x-x^2}}{2} + \frac{25}{2} \sin^{-1} \left(\frac{x-5}{5} \right) - \frac{2\sqrt{5}}{3} [x^{3/2}] \right]_0^5$$

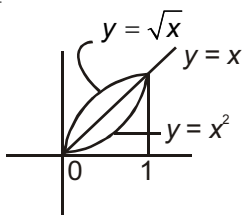
$$= 25 \left(\frac{\pi}{4} - \frac{2}{3} \right)$$

SECTION - F

Integer Answer Type Questions

1. If the area bounded by $y = x^2$ and $x = y^2$ is $\frac{k}{15}$ sq. units, then k is _____.

Sol. Answer (5)



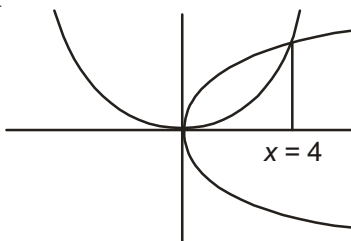
$$\frac{k}{15} = 2 \int_0^1 (x - x^2) dx = 2 \left(\frac{x^2}{2} - \frac{x^3}{3} \right)_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}$$

$$\Rightarrow k = \frac{15}{3} = 5 \quad k = 5$$

2. The area bounded by $y^2 = 4x$ and $x^2 = 4y$ is A_1 and the area bounded by $x^2 = 4y$, $x = 4$ and x -axis is A_2 . If $A_1 : A_2 = K : 1$ then K is _____.

Sol. Answer (1)

$$\begin{aligned} A_1 &= \int_0^4 \left(2\sqrt{x} - \frac{x^2}{4} \right) dx \\ &= \left(\frac{4}{3} x^{3/2} - \frac{x^3}{12} \right) \Big|_0^4 \\ &= \frac{4}{3}(8) - \frac{1}{12}(64) \\ &= \frac{16}{3} \text{ sq. units} \end{aligned}$$



$$\begin{aligned} A_2 &= \int_0^4 \frac{x^2}{4} dx \\ &= \left[\frac{x^3}{12} \right]_0^4 = \frac{16}{3} \end{aligned}$$

$$\Rightarrow A_1 : A_2 = 1 : 1$$

3. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \leq d_1(P) + d_2(P) \leq 4$, is

[JEE(Advanced)-2014]

Sol. Answer (6)

$$x > 0, y > 0, \quad 2 \leq \left| \frac{x-y}{\sqrt{2}} \right| + \left| \frac{x+y}{\sqrt{2}} \right| \leq 4$$

$$\text{case (1), } x > y \quad 2 \leq \frac{x-y}{\sqrt{2}} + \frac{x+y}{\sqrt{2}} \leq 4$$

$$2 \leq \frac{2x}{\sqrt{2}} \leq 4$$

$$1 \leq \frac{x}{\sqrt{2}} \leq 2$$

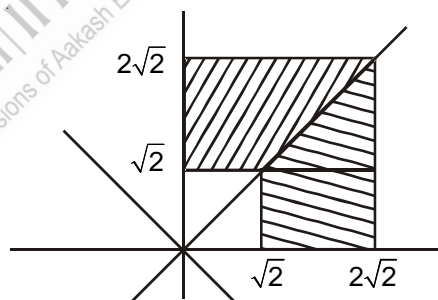
$$\sqrt{2} \leq x \leq 2\sqrt{2} \quad \dots(i)$$

Similarly $y > x$

$$\sqrt{2} \leq y \leq 2\sqrt{2} \quad \dots(ii)$$

Required area = area of shaded region

$$\begin{aligned} &= 2 \left(\sqrt{2} \times \sqrt{2} + \frac{1}{2} \times \sqrt{2} \times \sqrt{2} \right) \\ &= 2(2 + 1) = 6 \text{ sq. unit.} \end{aligned}$$



4. A farmer F_1 has a land in the shape of a triangle with vertices at $P(0, 0)$, $Q(1, 1)$ and $R(2, 0)$. From this land, a neighbouring farmer F_2 takes away the region which lies between the side PQ and a curve of the form $y = x^n$ ($n > 1$). If the area of the region taken away by the farmer F_2 is exactly 30% of the area of $\triangle PQR$, then the value of n is _____. [JEE(Advanced)-2018]

Sol. Answer (4)

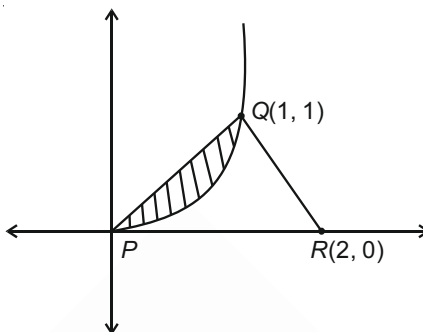
$$\text{Area of } \triangle PQR = \frac{1}{2} (2)(1) = 1$$

$$0.3 = \int_0^1 (x - x^n) dx$$

$$\Rightarrow \frac{3}{10} = \left[\frac{x^2}{2} - \frac{x^{n+1}}{n+1} \right]_0^1$$

$$\Rightarrow \frac{1}{n+1} = \frac{1}{2} - \frac{3}{10}$$

$$\Rightarrow n = 4$$

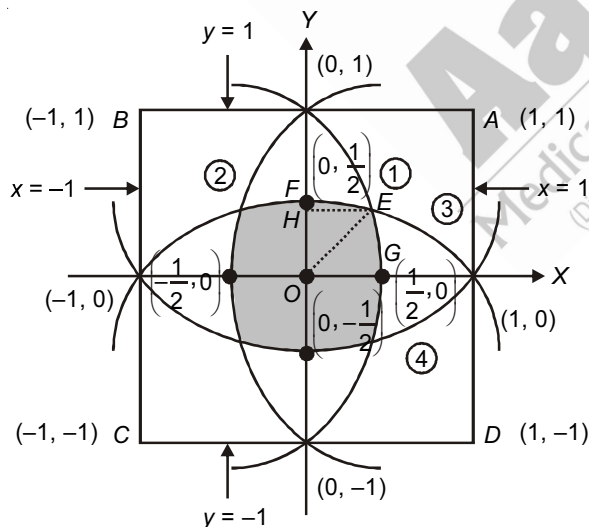


SECTION - G

Aakash Challengers Questions

1. Consider a square with vertices at $(1, 1)$, $(-1, 1)$, $(-1, -1)$ and $(1, -1)$. Let S be the region consisting of all points inside the square which are nearer to the origin than to any edge. Sketch the region S and find its area.

Sol.



Let (x, y) be any point inside the region S . Then from the hypothesis of the problem

$$\sqrt{x^2 + y^2} < |1-x|, |1+x|, |1-y|, |1+y|$$

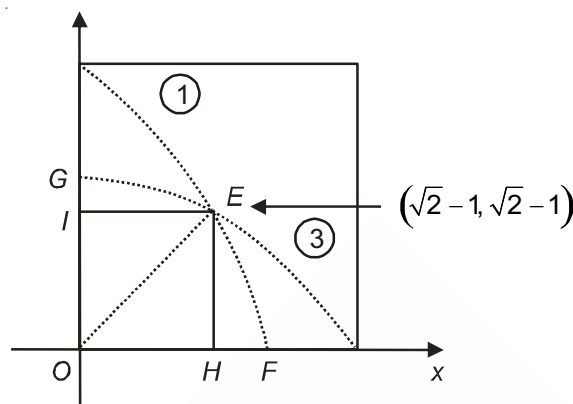
$$\Rightarrow x^2 + y^2 < (1-x)^2, (1+x)^2, (1-y)^2, (1+y)^2$$

$$\Rightarrow y^2 < 1 - 2x, 1 + 2x, x^2 < 1 - 2y, \text{ and } x^2 < 1 + 2y$$

The equations $y^2 = 1 - 2x$, $y^2 = 1 + 2x$, $x^2 = 1 - 2y$ and $x^2 = 2y + 1$ represent parabolas with vertices

$$\left(\frac{1}{2}, 0\right), \left(-\frac{1}{2}, 0\right), \left(0, \frac{1}{2}\right) \text{ and } \left(0, -\frac{1}{2}\right).$$

The required area = 4 times the area in the first quadrant. Let us just draw the area in the first quadrant



The area GEFOG = 2 area OEFO = 2(area DOEH + area HEFH)

$$= 2 \cdot \frac{1}{2} (\sqrt{2}-1)^2 + 2 \int_{\sqrt{2}-1}^{\frac{1}{2}} \sqrt{1-2x} \cdot dx$$

$$= (\sqrt{2}-1)^2 + 2 \cdot \left[\frac{2(1-2x)^{\frac{3}{2}}}{3(-2)} \right]_{\sqrt{2}-1}^{\frac{1}{2}}$$

$$= 2 + 1 - 2\sqrt{2} + \frac{2}{3} (1 - 2(\sqrt{2}-1))^{\frac{3}{2}}$$

$$= 3 - 2\sqrt{2} + \frac{2}{3} ((\sqrt{2}-1)^2)^{\frac{3}{2}}$$

$$= 3 - 2\sqrt{2} + \frac{2}{3} \{2\sqrt{2} - 1 - 3\sqrt{2}(\sqrt{2}-1)\}$$

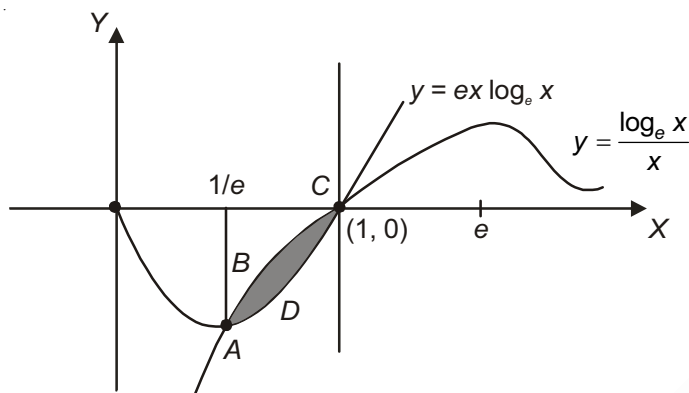
$$= 3 - 2\sqrt{2} + \frac{2}{3} (5\sqrt{2} - 7)$$

$$= \frac{1}{3} (4\sqrt{2} - 5)$$

$$\text{Required area} = \frac{4}{3} (4\sqrt{2} - 5) \text{ sq. units}$$

2. Evaluate the area of the region bounded by the curves $y = ex \log_e x$ and $y = \frac{\log_e x}{ex}$.

Sol.



Given curves are

$$y = ex \log_e x \quad \dots(i)$$

$$\text{and } y = \frac{\log_e x}{ex} \quad \dots(ii)$$

We have to find the points of intersection of (i) and (ii) of curves.

$$\text{From (i) and (ii), } ex \log_e x = \frac{\log_e x}{ex}$$

$$\Rightarrow \left(ex - \frac{1}{ex} \right) \log_e x = 0$$

$$\Rightarrow \log_e x = 0 \text{ or } ex - \frac{1}{ex} = 0$$

$$\Rightarrow x = 1 \text{ or } x^2 = \frac{1}{e^2}$$

$$\Rightarrow x = 1 \text{ or } x = \frac{1}{e} \quad (\because x > 0)$$

The required area = shaded area ABCDA

$$\begin{aligned} &= \int_{\frac{1}{e}}^1 (y_1 - y_2) dx \\ &= \int_{\frac{1}{e}}^1 \left[\frac{\log_e x}{ex} - ex \log_e x \right] dx \\ &= \frac{1}{e} \int_{\frac{1}{e}}^1 \frac{\log_e x}{x} dx - e \int_{\frac{1}{e}}^1 x \log_e x dx \\ &= \frac{1}{e} I_1 - e I_2 \quad \dots(iii) \end{aligned}$$

$$\begin{aligned}
 \text{Now } I_1 &= \int_{\frac{1}{e}}^1 \log x \cdot \left(\frac{1}{x}\right) dx \\
 &= \left[\frac{(\log x)^2}{2} \right]_{\frac{1}{e}}^1 \\
 &= \frac{1}{2} \left[0 - \left(\log \frac{1}{e} \right)^2 \right] \\
 &= \frac{1}{2} (-1) = -\frac{1}{2} \quad \dots(\text{iv})
 \end{aligned}$$

$$\begin{aligned}
 \text{and } I_2 &= \int_{\frac{1}{e}}^1 x \cdot \log x \, dx \\
 &= \left[\log x \cdot \frac{x^2}{2} \right]_{\frac{1}{e}}^1 - \int_{\frac{1}{e}}^1 \frac{1}{x} \cdot \frac{x^2}{2} \, dx \\
 &= 0 - \log \frac{1}{e} \cdot \left(\frac{1}{2e^2} \right) - \left[\frac{x^2}{4} \right]_{\frac{1}{e}}^1 \\
 &= \frac{1}{2e^2} - \frac{1}{4} + \frac{1}{4e^2} \\
 &= -\frac{1}{4} + \frac{3}{4e^2} \quad \dots(\text{v})
 \end{aligned}$$

From (iii), (iv) and (v), we have

$$\begin{aligned}
 \text{Required area} &= -\frac{1}{2e} + \frac{e}{4} - \frac{3}{4e} \\
 &= \frac{e}{4} - \frac{5}{4e} \\
 &= \frac{e^2 - 5}{4e} \text{ sq. units}
 \end{aligned}$$

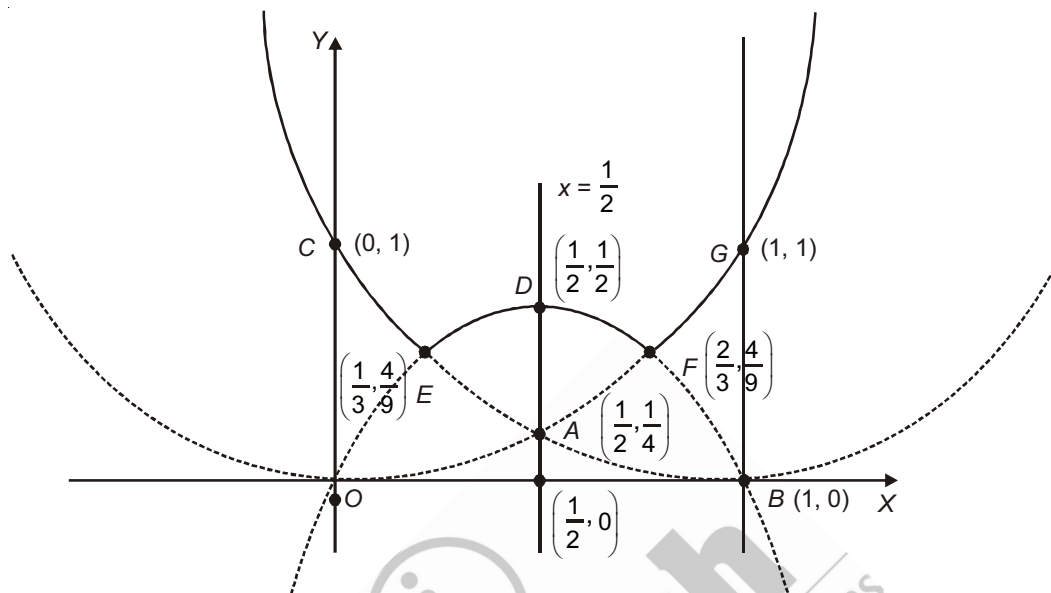
3. Let $f(x) = \max\{x^2, (1-x)^2, 2x(1-x)\}$, where $0 \leq x \leq 1$. Determine the area of the region bounded by the curves $y = f(x)$, x -axis, $x = 0$ and $x = 1$.

Sol. First we draw the graph of curves

$$y = x^2 \text{ or } x^2 = y \quad \dots(\text{i})$$

$$y = (1-x)^2 \text{ or } (x-1)^2 = y \quad \dots(\text{ii})$$

and $y = 2x(1-x)$ or $\left(x - \frac{1}{2}\right)^2 = -\frac{1}{2}\left(y - \frac{1}{2}\right)$... (iii)



From the graph it turns out that

$$y = \begin{cases} (1-x)^2 & ; 0 \leq x \leq \frac{1}{3} \\ 2x(1-x) & ; \frac{1}{3} \leq x \leq \frac{2}{3} \\ x^2 & ; \frac{2}{3} \leq x \leq 1 \end{cases}$$

Required area OCEDFGBO

$$\begin{aligned} &= \int_0^{\frac{1}{3}} (1-x)^2 dx + \int_{\frac{1}{3}}^{\frac{2}{3}} (2x-2x^2) dx + \int_{\frac{2}{3}}^1 x^2 dx \\ &= \left[\frac{(1-x)^3}{-3} \right]_0^{\frac{1}{3}} + \left[x^2 - \frac{2x^3}{3} \right]_{\frac{1}{3}}^{\frac{2}{3}} + \left[\frac{x^3}{3} \right]_{\frac{2}{3}}^1 \\ &= \left[-\frac{8}{81} + \frac{1}{3} \right] + \left[\left(\frac{4}{9} - \frac{16}{81} \right) - \left(\frac{1}{9} - \frac{2}{81} \right) \right] + \left[\frac{1}{3} - \frac{8}{81} \right] \\ &= -\frac{8}{81} + \frac{1}{3} + \frac{1}{3} - \frac{14}{81} + \frac{1}{3} - \frac{8}{81} \\ &= 1 - \frac{30}{81} = \frac{51}{81} = \frac{17}{27} \text{ sq. units} \end{aligned}$$

4. Find the area enclosed by circle $x^2 + y^2 = 4$, parabola $y = x^2 + x + 1$, the curve $y = \left\lceil \sin \frac{x}{4} + \left\lceil \cos \frac{x}{4} \right\rceil \right\rceil$ and x-axis (where $[\cdot]$ is the greatest integer function)

Sol. $y = \left\lceil \sin \frac{x}{4} + \left\lceil \cos \frac{x}{4} \right\rceil \right\rceil$

$$1 \leq \left\lceil \sin \frac{x}{4} + \left\lceil \cos \frac{x}{4} \right\rceil \right\rceil \leq \sqrt{2}$$

$$\Rightarrow y = 1$$

Now we have to find the area enclosed by the circle $x^2 + y^2 = 4$, parabola

$$y - \frac{3}{4} = \left(x + \frac{1}{2}\right)^2, \text{ line } y = 1 \text{ and x-axis}$$

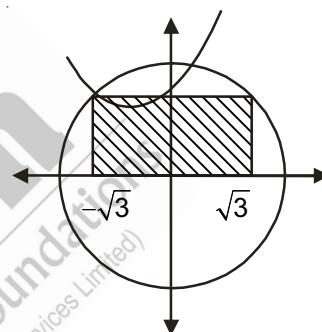
Hence the required area

$$= \sqrt{3} \times 1 + (\sqrt{3} - 1) \times 1 + \int_{-1}^0 (x^2 + x + 1) dx + 2 \int_{\sqrt{3}}^2 \left(\sqrt{4 - x^2} \right) dx$$

$$= (2\sqrt{3} - 1) + \left[\frac{x^3}{3} + \frac{x^2}{2} + x \right]_{-1}^0 + 2 \left[\frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) \right]_{\sqrt{3}}^2$$

$$= (2\sqrt{3} - 1) + \frac{5}{6} + \frac{2\pi}{3} - \sqrt{3}$$

$$= \left(\frac{2\pi}{3} + \sqrt{3} - \frac{1}{6} \right) \text{ sq. unit}$$



5. Let $f(x) = \max \left\{ \sin x, \cos x, \frac{1}{2} \right\}$, then area of the region bounded by the curves $y = f(x)$, x-axis, y-axis and $x = 2\pi$.

Sol. $f(x) = \max \left\{ \sin x, \cos x, \frac{1}{2} \right\}$

$$I = \int_0^{\frac{\pi}{4}} \cos x dx + \int_{\frac{\pi}{4}}^{\frac{5\pi}{6}} \sin x dx + \int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} \frac{1}{2} dx + \int_{\frac{3\pi}{2}}^{2\pi} \cos x dx = \left(\frac{5\pi}{12} + \sqrt{2} + \sqrt{3} \right) \text{ sq. units}$$

